

HK IMO Training 2020-2021
Problem Set 18
For 13 March 2021



Problem 69. The edges of a complete graph on n vertices, where $n \geq 3$, are coloured with n colours, and every colour appears at least once. Prove that one can find a triangle whose sides are coloured with 3 different colours.

Problem 70. If the number of edges of a simple undirected graph on n vertices is at least $\frac{n^2}{4} + 2$, where $n \geq 5$, show that there exist 2 triangles which have exactly one common vertex.

Problem 71. If the number of edges of a simple undirected graph on n vertices is at least $\frac{n\sqrt{n}}{2} + \frac{n}{4}$, prove that the graph contains a cycle of length 4.

Problem 72. Prove the following statements:

- (a) If a simple undirected graph on n vertices does not contain a complete bipartite graph with 2 vertices on one vertex class and 3 vertices on the other vertex class, then its number of edges is at most $\frac{n\sqrt{n}}{\sqrt{2}} + n$.
- (b) Given $n \geq 16$ distinct points $P_1, P_2, \dots, P_n$ in the plane, then at most $n\sqrt{n}$ of the segments P_iP_j have unit length.